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EST II — MATH LEVEL 2 • UNIT 2.01

Sequences, Series & Sigma Notation

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Sequences, Series & Sigma Notation Practice Exercises

1 MCQ

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

A 14

B 17

C 8

D 11

2 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

A 96

B 192

C 98

D -96

3 MCQ

Consider a geometric sequence with positive common ratio $r > 0$ where the first term is $a_1 = \frac{1}{2}$. If the sum of the first three terms is $S_3 = \frac{7}{2}$, find the value of r.

A 2

B -2

C 1/2

D 3

4 MCQ

A geometric sequence has $a_1 = 2$ and common ratio $r = 3$. Find a_5.

A 162

B 486

C 165

D -162

5 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 3$. Find a_5.

A 486

B 1458

C 489

D -486

6 MCQ

A geometric sequence has $a_1 = 2$ and common ratio $r = -2$. Find a_4.

A -16
B 32
C -18
D 16

7 MCQ

A geometric sequence has $a_1 = 5$ and common ratio $r = 3$. Find a_3.

A 45
B 135
C 48
D -45

8 MCQ

A geometric sequence has $a_1 = 1$ and common ratio $r = -3$. Find a_5.

A 81
B -243
C 78
D -81

9 MCQ

In an applied problem, a geometric sequence has $a_1 = 6$ and common ratio $r = 3$. Find a_5.

A 1458
B 489
C -486
D 486

10 MCQ

Suppose a geometric sequence has $a_1 = 5$ and common ratio $r = 3$. Find a_3.

A 135
B 48
C -45
D 45

11 MCQ

In an applied problem, a geometric sequence has $a_1 = 1$ and common ratio $r = -3$. Find a_6.

A -243
B -246
C 243
D 729

12 MCQ

An arithmetic sequence has its first term $a_1 = 7$ and a common difference $d = 5$. What is the value of the 3rd term, a_3?

A 12
B 17
C 21
D 35

13 MCQ

Suppose a geometric sequence has $a_1 = 1$ and common ratio $r = -3$. Find a_4.

A 27
B 81
C -30
D -27

14 MCQ

Find the common difference of the arithmetic sequence $6,\ 4,\ 2,\ 0, \ldots$

A -2
B 4
C 2
D -1

15 MCQ

Find the common difference of the arithmetic sequence $-5,\ 3,\ 11,\ 19, \ldots$

A 8
B -8
C 9
D 3

16 MCQ

Find the common difference of the arithmetic sequence $-2,\ -8,\ -14,\ -20, \ldots$

A 6
B -5
C -6
D -8

17 MCQ

Calculate the common difference of the arithmetic sequence $6,\ 11,\ 16,\ 21, \ldots$

A 5
B -5
C 6
D 11

18 MCQ

A sequence is defined by the recurrence relation $a_{n+1} = 2a_n - 3$ for $n \geq 1$, with $a_1 = 5$. Consider a second sequence, b_n, where $b_n = a_n - k$ for some constant k. If the sequence b_n is a geometric sequence, find the value of k and the common ratio of the sequence b_n.

A k = 2, r = 3

B k = 3, r = 2

C k = 3, r = 3

D k = -3, r = 2

19 MCQ

Consider the arithmetic sequence shown on the number line. What is the common difference of this sequence?



A -3

B 12

C 6

D 3

20 MCQ

Find the common ratio of the geometric sequence $5,\ 10,\ 20,\ 40, \ldots$

A -2

B 3

C 5

D 2

21 MCQ

Find the common ratio of the geometric sequence $5,\ -10,\ 20,\ -40, \ldots$

A 2

B -2

C -1

D -15

22 MCQ

Calculate the common ratio of the geometric sequence $4,\ -8,\ 16,\ -32, \ldots$

A -2

B 2

C -1

D -12

23 MCQ

A geometric sequence has its first term a_1 and a common ratio r. The sum of the first three terms is 26, and the sum of the first six terms is 728. If $a_1 = 2$, find the value of r.

A 2

B -2

C -3

D 3

24 MCQ

Calculate the common ratio of the geometric sequence $1,\ 4,\ 16,\ 64, \ldots$

A -4

B 4

C 5

D 3

25 MCQ

Find the sum of the first 6 terms of the arithmetic sequence with $a_1 = 9$ and $d = 3$.

A 105

B 99

C 144

D 96